



Adhesion Layer Optimization for Parylene F Encapsulated Yttrium Foils

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ABSTRACT

- We are creating a non-invasive parylene F yttrium radiation patch treatment for Basal Cell Carcinomas (skin cancers)
- We used an AJA Ultra High Multi-Technique Deposition System to deposit 50 nm yttrium and aluminum thin films and coat Silicon Wafers
- We applied an adhesion promotor, [3-(Methacryloxy)propyl]trimethoxysilane (A-174 Silane), to the coated wafers using 3 methods; A quick dip in solution, a long dip in solution and vapor deposition
- We used a Nicolet 6700 Fourier Transform Infrared Spectrometer (FTIR) to compare the chemical bonds formed between A-174 silane, yttrium and aluminum for the 3 methods

Treatments and why use an yttrium patch (Y)

- Current treatments, like skin grafting surgery and x-ray or proton radiation therapy, can leave patients disfigured, or exposed to high energy particles
$$^{89}\text{Y} + \text{neutron} \rightarrow ^{90}\text{Y}$$

$$^{90}\text{Y} \rightarrow ^{90}\text{Zr} + \beta^-$$
- When yttrium is irradiated in a TRIGA reactor, it forms Y-90 and undergoes radioactive decay and emits a beta electron with a short 2 ½ day half-life [1]
- Beta electrons only penetrate skin on the order of 2 mm with a max energy of (2.2 MeV), and drop of quickly compared to x-rays or gamma rays
- Basal Carcinomas exist (1-3 mm) below the surface of the skin depending on severity

Parylene F Cover + Adhesive A174 Silane

- Irradiated yttrium needs to be encapsulated to avoid radiation travel to other body areas
- A coating of the biomedical substance parylene F, which protects from moisture, corrosion, salt, solvents, and contaminants, can be applied via tumbling but does not bond easily to metals
- We use the adhesion promoter [3-(Methacryloyloxy)propyl]trimethoxysilane (A-174 Silane) to bond with yttrium/aluminum – OH oxides on the surface
- Silane has a chemically unstable C=C bond that will react with parylene F to link up

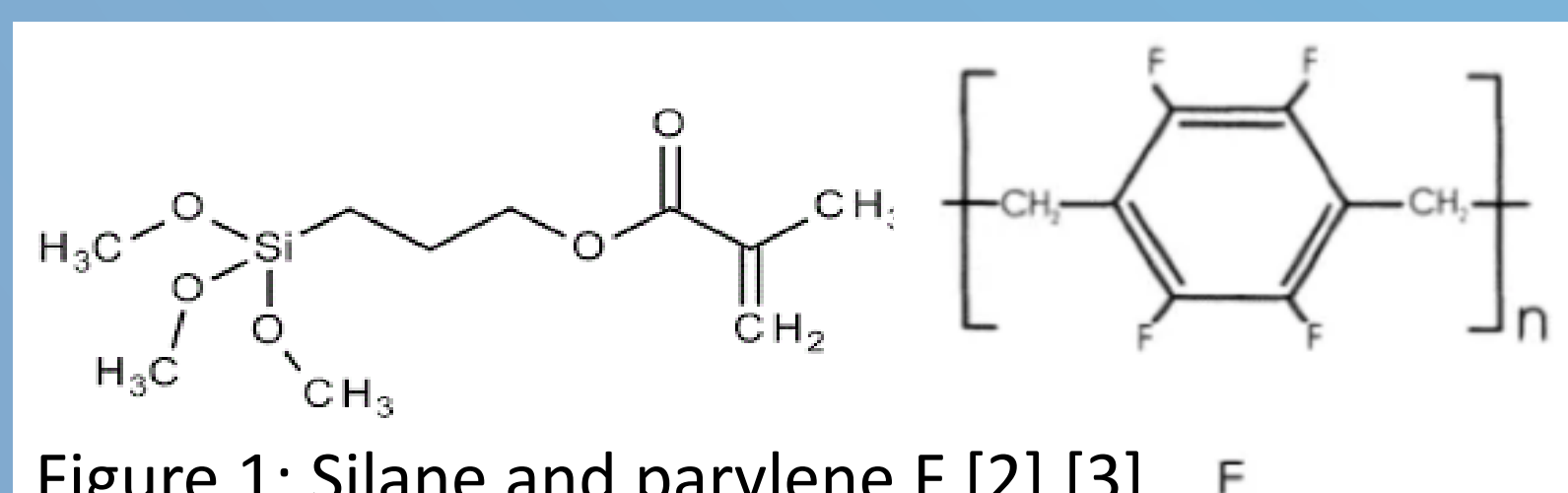


Figure 1: Silane and parylene F [2] [3]

A-174 Silane Coating Experimental Methods

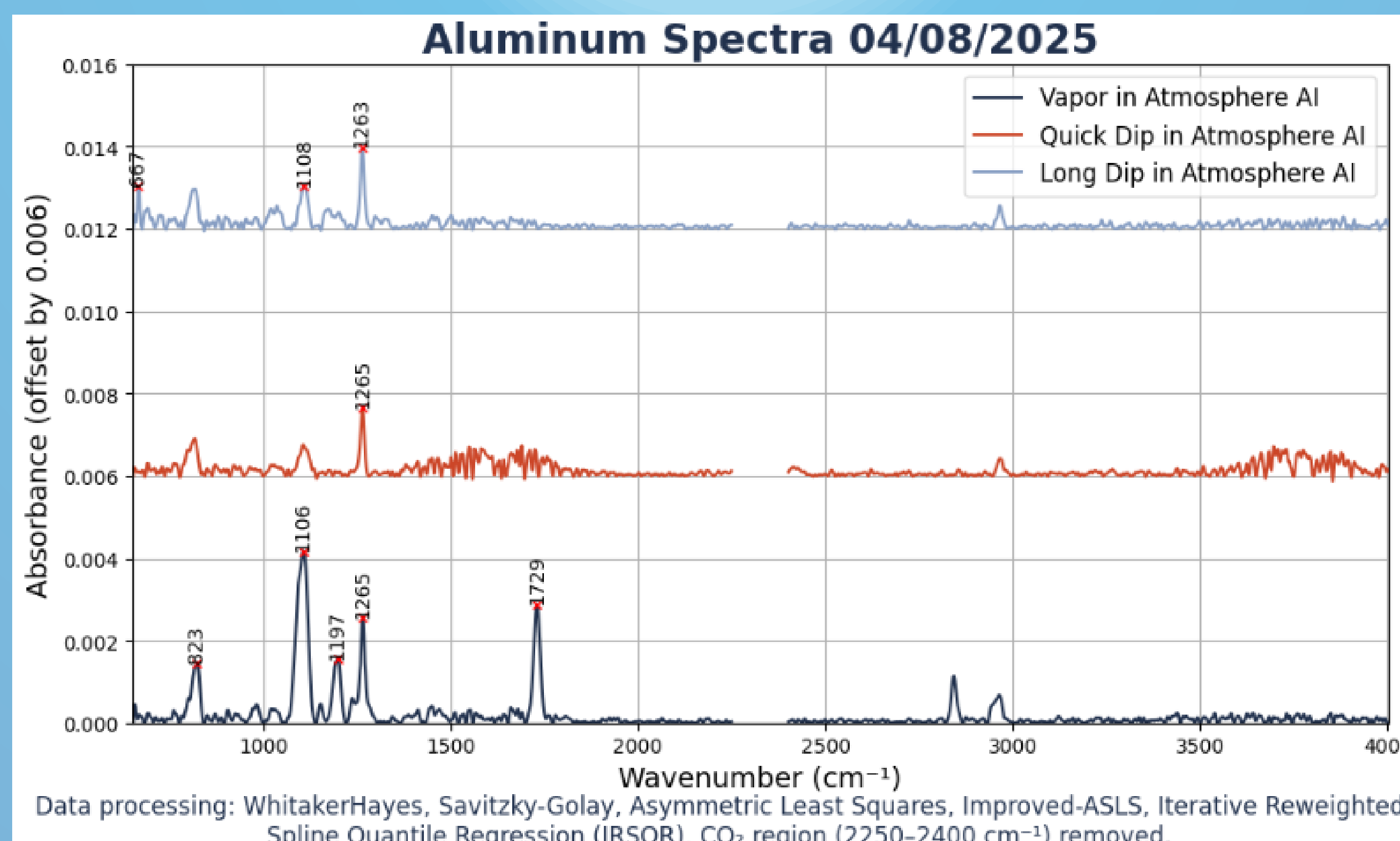
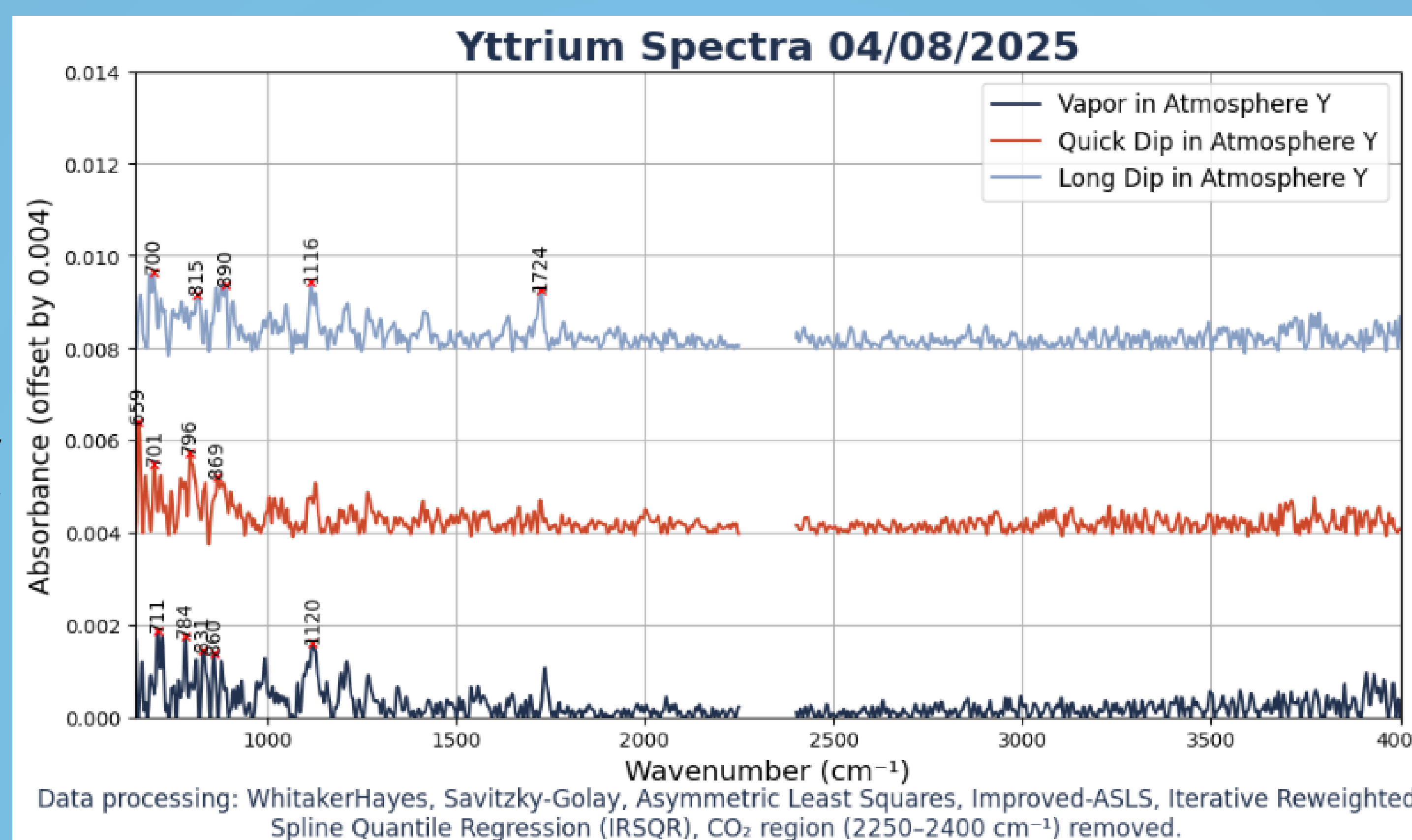
- A-174 Silane coats has been applied to aluminum in literature, so we used aluminum to compare to Yttrium
- A174 Silane (<1%) is dissolved in a solution of isopropanol + water
- Samples are placed in the solution for a quick dip (few seconds) or a long dip (30 minutes), allowed to air dry, and then are baked
- When water reacts with yttrium it forms H₂ bubbles, and Y₂O₃ oxide builds up which we do not want
- We took data on a vapor deposition technique performed by VSI Parylene per their specifications without drying or baking

Data Collected Settings Fourier Transform Infrared Spectroscopy (FTIR) Nicolet 6700

- We collected data with the FTIR at specific settings
- Number of scans 128
- Resolution of 4/cm
- A moving mirror speed of 1.27 cm/sec
- A gain of 8

Engineering Constraint/Standards

- The best adhesion layer application process is done in a water-containing liquid
- This liquid will oxidize the Yttrium, and is therefore not an option
- United States Nuclear Regulatory Commission: Sealed Biomedical Radiation Device must not emit > 185 Bq using wipe leak kit [4]



Data Analysis

- Pure A-174 Silane Spectra has bonds in cm⁻¹ at [5]
- 1078 cm⁻¹, Si-O-C asymmetric (asym) stretching: disappears during hydrolysis
- 850, 910 cm⁻¹ are Si-OH vibration modes; silanols
- 660, 690, 884 cm⁻¹ represent Si-O-Al and Si-O-Y metal hydroxides and silanols
- 1015 cm⁻¹ shows Si-O-Si asym
- 1717 cm⁻¹ shows C=O Methacrylate stretching
- 1161, 2945 cm⁻¹ CH₃ bending and C-H st (CH₃)
- 1296 cm⁻¹ is C-O symmetric stretch in Methacrylate

Results

- Two peaks at 2339, and 2370 cm⁻¹ in both sets of data represent O=C=O or carbon dioxide stretching which is unimportant and commonly deleted in literature
- We found bond peaks to be lower in the quick dip and mirroring those in the long dip for yttrium at (700, 890, 1116, and 1724 cm⁻¹) except at 659 and 796 cm⁻¹ which were opposite
- The vapor method mimicked the long dip in yttrium, except for a higher peak at 831 cm⁻¹, compared to the long dip at 815 cm⁻¹
- We found in aluminum that the vapor method had higher peaks at 823, 1106, 1197, 1265, and 1729 cm⁻¹ than both dip methods except at 667 cm⁻¹
- A distinct peak is visible at 1729 cm⁻¹ in the vapor method for aluminum which is close to 1717 cm⁻¹ and shows C=O Methacrylate stretching, and is typically characteristic of any silane presence
- This peak is reduced by 0.002 in the quick dip, and 0.0025 in the long dip which seems to represent deformation of the C=O bond as it interacts with water

Conclusions/Future Work

- The existence of a peak found in the long dip method located at 890 cm⁻¹ and 659 cm⁻¹ in the quick dip with an absorbance found within 0.001 of the vapor deposition for yttrium suggest that although peaks for A174-Silane-yttrium bonds exist, lower sample reflectivity led to a lower and noisier signal than in the aluminum samples
- Results above indicate a need to refine the process for vapor deposition of adhesion promoter [3-(Methacryloyloxy)propyl]trimethoxysilane on both yttrium and aluminum

References

- [1] National Center for Biotechnology Information. PubChem Compound Summary for CID 23993, Yttrium. Accessed Oct 9, 2024.
- [2] "PARYLENE CHEMICAL PROPERTIES: PRODUCT PROTECTION THAT MAY SURPRISE YOU", VSI PARYLENE, © 2025, Vertical Solutions, Inc., Accessed 10/7/24.
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- [4] "39.35 Leak Testing of sealed sources.", United States Nuclear Regulatory Commission, Tuesday, Aug 29, 2017.
- [5] Pantoja, M, Diaz-Benito, B, et.al, "Analysis of hydrolysis process of γ-methacryloxypropyltrimethoxysilane and its influence on... 6063 aluminum alloy", *Applied Surface Science*, vol 255, issue 12, April 1, 2009. SN 0169-4332.

Acknowledgements

- Thanks to Alex Dixon, Tom Furtak, and Molly Brown
- This work funded in part by the Colorado Office of Economic Department



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